

Final Report

Team Gamma

Romano Almaza

Ahmed Badv

Brendan S

Ryan Thom

Abstract

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Introduction

Team Charter

Our team charter consisted of expectations, goals, and norms. Together these expectations, goals, and norms have allowed us to be productive despite the heavy course load we are all enduring. One of our big expectations for this group was that everyone would carry their own weight and play their role in order to maximize the quality of the potential outcome. Another main goal/expectation was communication, we wanted communication between members to be versatile in order to get as many perspectives as possible and have good organization and time management. A main obstacle that we were able to anticipate was time management and schedule planning due to the fact that everyone has their own schedules and work to tend to. Some of our main norms fall under the category of work, meeting, communication, and consideration norms:

Work norms:

Our work norms consisted of trying to spend at least 6 hours a week working on this project, understanding each other's strengths and weaknesses and assigning roles based on this to maximize efficiency. taking all options, perspectives, and opinions into consideration to find the best solution, setting goals in order to organize and schedule ourselves, set due dates for phases in our project for efficient time management

Meeting norms:

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Consideration norms for us were to keep an open mind to all suggestions, options and solutions.

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Product vision statement:

Our vision for the product we will be building is to help people who are physically limited to play fetch with their pets, enjoy their pet's company and strengthen the owner-pet bond. With this machine, we hope to improve the life of dog owners and dogs themselves by creating memorable moments and memories between them.

Project specifications:

Due to us having a very specific vision for our product's present and future, we came up with specifications that we wanted to integrate into our product to help us reach our vision, these being:

- The product must have a safety mechanism in future to prevent injuries to the dog and owner when the ball is launched
- The product must be portable to allow for easy transportation
- Well-balanced, cannot be knocked over easily
- Simple and easy to use for owners and dogs, user friendly
- Must be modular, to allow for future upgrades and improvements
- The power source must be affordable, easily accessible, and allow for the machine to still be portable (batteries)

Metrics:

In the early stages of our product, we projected some early metrics that acted as goals of what we wanted to achieve through our product. Initially, the preferred weight for our product was between 2kg and no more than 4kg. Currently, the main segment of our product is around 2kgs so we are possibly looking at getting close to 4kgs when the project is 100% completed. We initially planned for our product to be 30-40 cm all around (cube), but as of right now, it's looking like it will be bigger. Our current product is bigger and heavier than we expected it to be, this is only a temporary issue as once we have our product fully functional and working the exact way we visioned we can look to deduce and make adjustments in the portability area. Our goal projectile distance is 20-30 ft, our most recent test saw the ball only shooting around 8-9 feet (rough estimate), this is because our flywheel diameter wasn't big enough and we had to add material to it to increase the diameter putting the flywheels off balance and lowering the power, velocity, and accuracy with which the ball was shot at. Our solution to this was very easy as we just reprinted bigger flywheels, and this also fixed the centring problem we had.

Limitations:

Some limitations we ran into were budgeting, it was hard to find the parts we needed at a reasonable price without cheapening out on quality and performance. Along with this came the time it took for the parts to ship since we could only find them online, this pushed back our full assembly for a bit. We were also limited by 3D printing times as many of our pieces took days (one took 3) to print in order to achieve our desired quality. Due to the fact that none of the people in our group owns dogs, our testing time and accessibility to dogs was very limited, though we had connections to people with dogs it meant that we would have to abide by their schedules in turn scrambling and forcing us to adjust our own schedules. We didn't have access to lots of equipment needed to build this prototype minus a 3d printer, like before we had connections to people with the equipment that we needed but again it meant working on their schedule. Due to not hearing back from our client, we lost about 3 weeks of crucial time putting us in a bit of a time crunch.

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5. Make the prototype modular to allow for future improvements, attachments, and fixes.
6. Flashlight/bright light source for the dark
7. Camera to monitor dog, the camera can also be configured to assist other mechanisms in the product like the safety mechanism

Interior components:

8. Funnel, to allow and make it easy for the dog to use the machine by itself
9. Automation, we plan to add code in order to make the product fully run by itself
10. The safety mechanism incorporates a sensor that detects anything within 1-2 ft of the nozzle and locks the entire machine if something is present
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12. Conveyer belt, so the ball doesn't hit the flywheel as soon as it dropped
13. Slope, allowing for the ball to roll into the flywheels with some speed and velocity already built up
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16. Magnetic plates to shoot the ball
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Concept screening

Ideas cut out: 1, 2, 12, 16, 18

Ideas used for concept matrix: 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 17, 19

Concept matrix:

Presentation of Top Ideas

Our top ideas for this product are as listed:

Flywheel:

The flywheel was our first and top concept shooting mechanism from the very beginning of the semester due to the fact that it is easily understandable, not complex (in case we need to make adjustments), and it has the potential for great speed and torque. The simplicity of the flywheel mechanism proved to be beneficial to us as we had to adjust the diameter of the flywheels multiple times throughout this project. This mechanism is essentially 2 wheels attached to motors on either side of the tube, one spinning clockwise and the other spinning counter clockwise, once the ball makes contact with the flywheels and pinches in between them the torque and rpm of the wheels are spinning at allow the flywheels to launch a ball at high speed and velocity.

Funnel:

One of the conceptual components that made our project stand out in our eyes is the funnel. Its main purpose was to allow the dog to put the ball into the machine by itself which in turn would lead to us having the option to completely automate the machine in the future so a dog could safely and independently use it. Incorporating the funnel would have led to us implementing a sensor and a stoppage mechanism so the ball could be shot at random intervals. In the end, though we had complete models ready to print, due to limitations with time, material, and budget, the funnel proved to be too costly in those terms and we ended up not using the funnel although we plan to integrate it in the future.

Automation:

Automation is currently a completely conceptual concept for us at the moment, we would like to include it in our product but due to the lack of time, we had we, unfortunately, had to let go of the idea for the time being. Our plan was stated previously in that we would have liked to have a sensor and something connected to the sensor that could stop the ball, then we would implement code that would release the ball at random intervals allowing it to hit the flywheel and be unpredictable so the dog won't get into a routine and still keep it excited to use the machine. This will also help in getting the dog to use the machine by itself.

Safety mechanism:

Along with the automation comes a safety mechanism to prevent injury to the dog or the owner. It will include a sensor and code that will turn off/ lock the shooting mechanism if

an object is detected within a foot or less of the nozzle. We aim to include this along with automation.

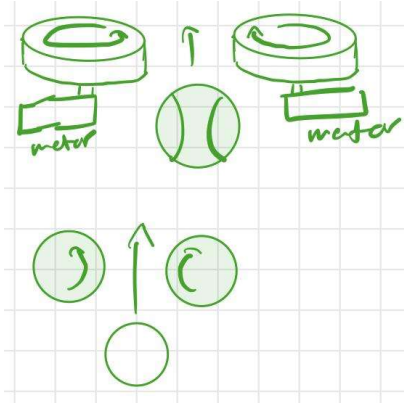
Concept Selection

Matrix of Top Ideas:

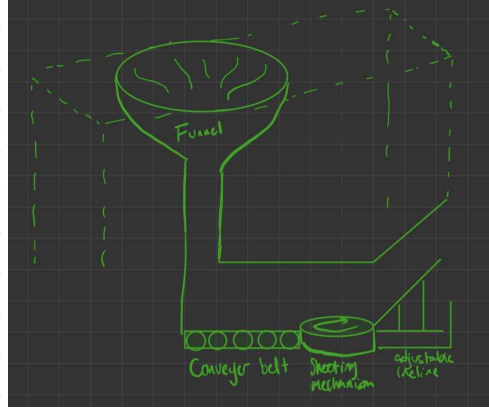
Justification:

Detailed Design

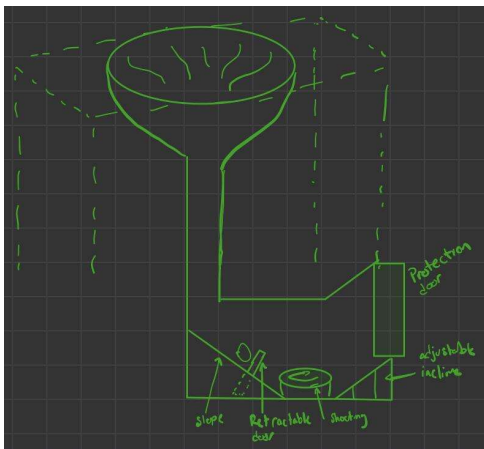
Schematics:



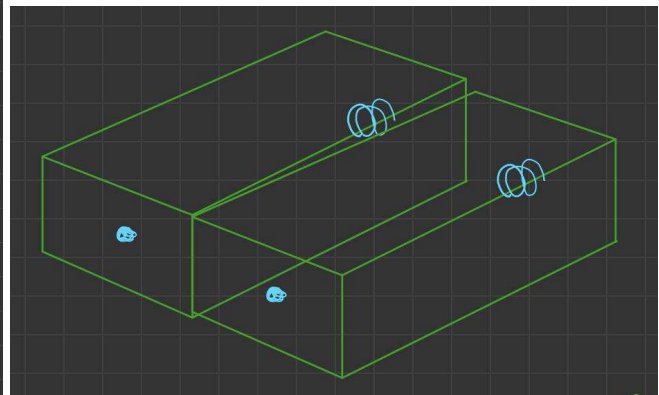
First Flywheel concept



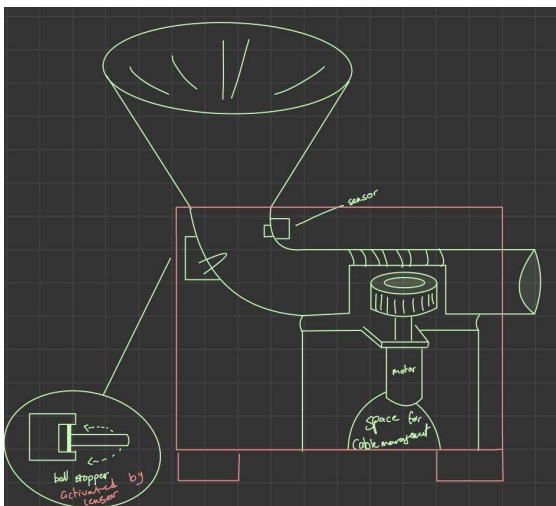
Concept with conveyor belt



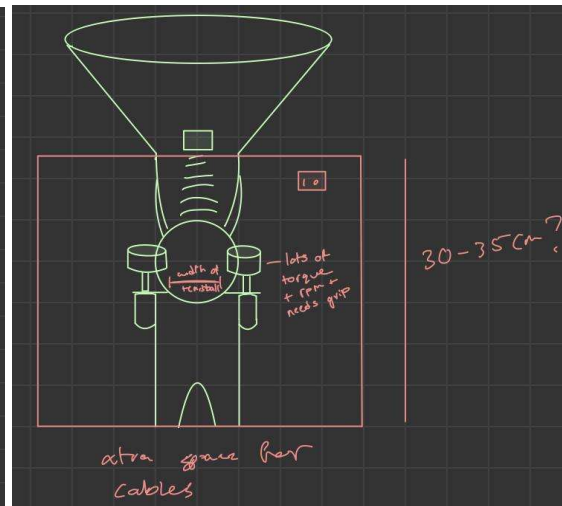
Concept with slope and stopper



Battery pack schematic

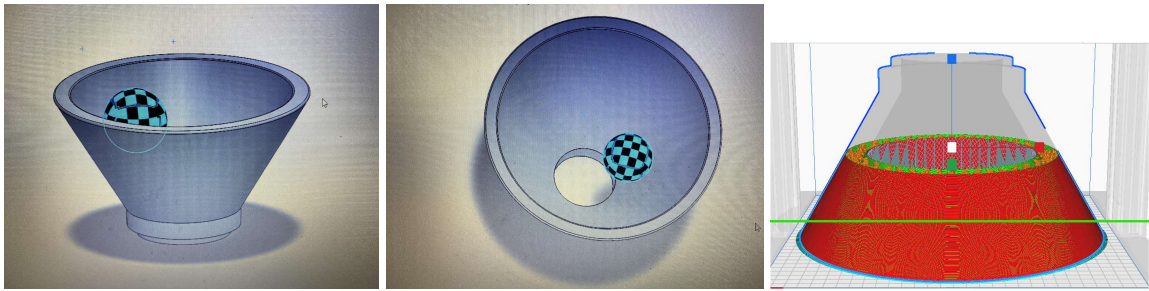


Final schematic side view

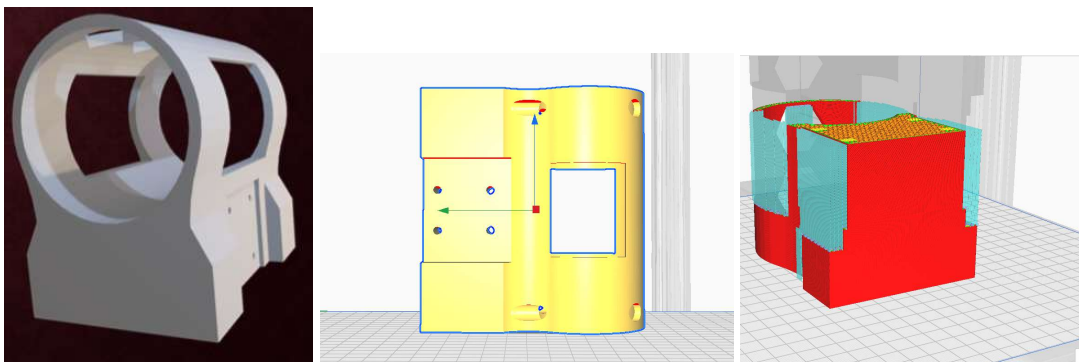


Final schematic front view

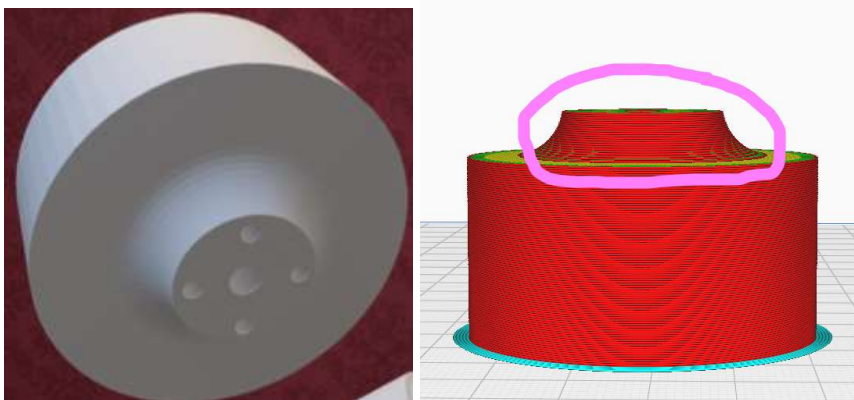
3D models/slicing:



3D models/slicing of the funnel



3D models/slicing of the shooting tube



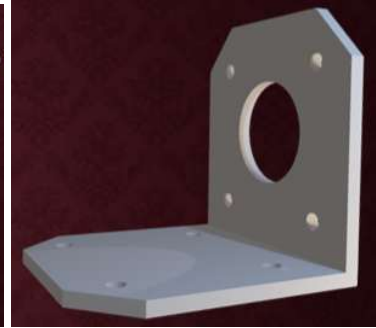
3d model/slicing of the flywheels



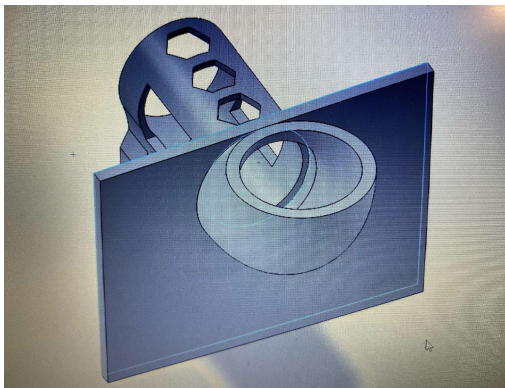
3D model of motor and flywheel



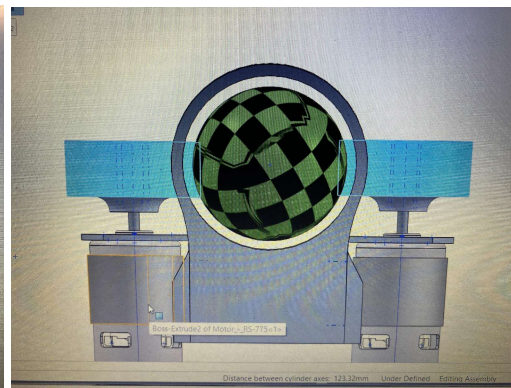
3D model of motor



3D model of the L bracket

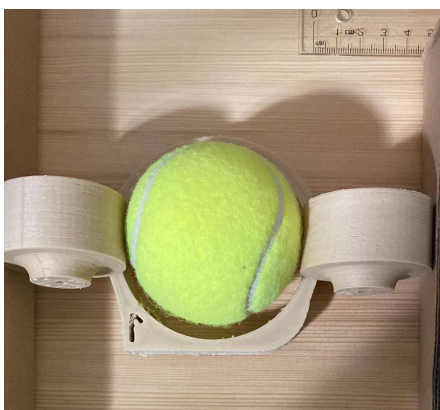


3D model of insert tube



3D model of everything put together

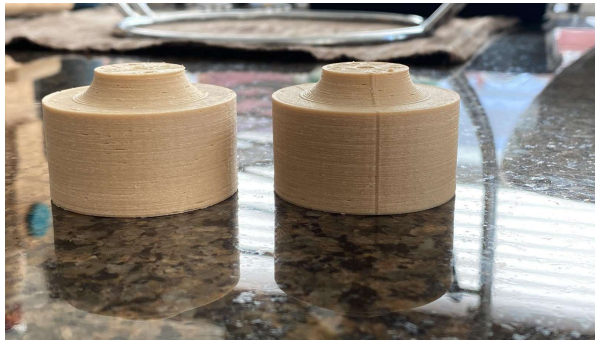
Final product:



Initial fitting test



Printing process



Initial flywheels



Flywheel and coupling



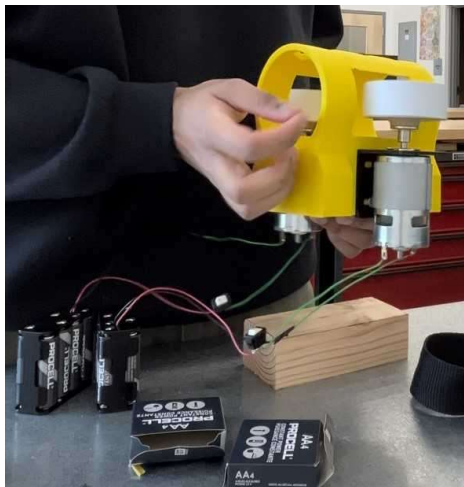
Final flywheels and motors



Partially assembled tube



Fitting test



Fully assembled tube with electrical components



Testing with dogs

Teamworking

As we are near the end of this project we were able to come up with a final version of our project management report:

Get Project Specifications

Before even starting a project having the specs is especially important. From a customer service perspective if you give the customer.

People involved:

Brendan

Generate Concept Solutions/designing

Before building our project we have to generate some concepts. The main idea is to think of problems we could encounter and find solutions to the problems. For example we need to throw a ball well how? During this process we will generate possible solutions to this problem.

People involved

Brendan, Romano, Ahmed, Ryan

Sketch final solution

After generating some concepts we have to narrow it down and find the one solution. This could be a combination of various components from our concepts, or could be something entirely different. The sketches should include drawings and schematics of the final product and every component within, with a listing specifics and notes. The final sketch will start to give us a direction on where to go and lead us to start the manufacturing process.

People involved

Ryan

Research materials

In order to build something you have to have the correct materials. Researching materials is crucial especially if you are on a budget. For example something like carbon fiber can get really expensive because you have to buy things like the fabric, epoxy and other minor things needed. We also have to find out which materials will be the best fit for our final product. This means if a material is cheap but is super fragile to work with it is probably not a good idea to use this material to make our prototype.

People involved

Brendan, Ryan

CAD Drawings

Using school provided software, Solidworks. We will design prototypes using this to have a formal blueprint of how we wish our product will look like.

People involved

Romano, Ahmed

Manufacturing

We need to build our product. This means booking shop times, getting access to special equipment. Getting access to resources. Managing resources and materials under budget. Quality assurance.

People involved.

Brendan, Romano, Ahmed, Ryan

Inspection

This is to make sure that components are up to par and address problems that may cause a disaster and fix them. This also make sure that there were no flaws during the manufacturing process like any cracks in parts

People involved

Ryan, Brendan, Romano

Testing

This is to make sure the product works and meets our standards and to ensure that it functions properly in real life scenarios rather than theoretical scenarios because theoretical results may differ from actual result (ie. tested with real dogs at a dog park)

People involved

Ryan



In summary our roles can be summarized as:

Brendan Sng: Resources, finance, communications, quality control/manufacturing (3D printing)

Ryan Thomas: Calculations, electrical, assembly, parts, testing

Romano Almazan: Solidworks/modeling, schematics, parts

Ahmed Badawy: Solidworks/modeling, schematics

We believe that our team was able to fairly distribute the work

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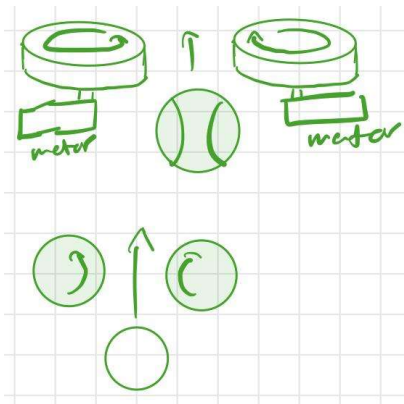
Concept Selection

Matrix of Top Ideas:

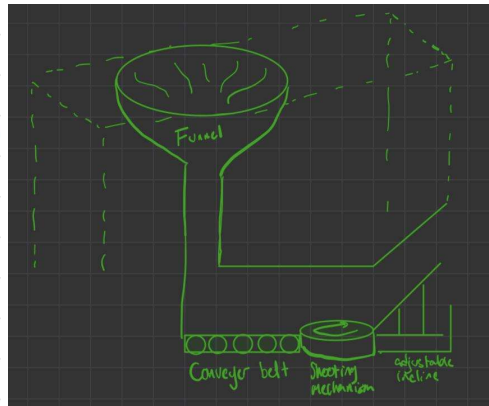
Justification:

Detailed Design

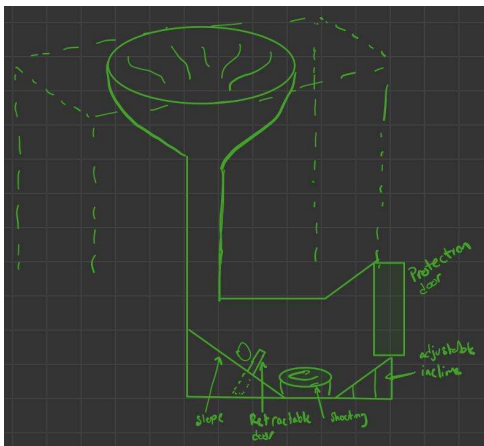
Schematics:



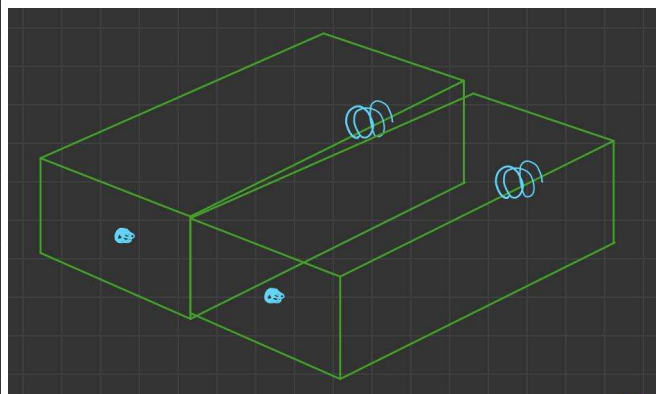
First Flywheel concept



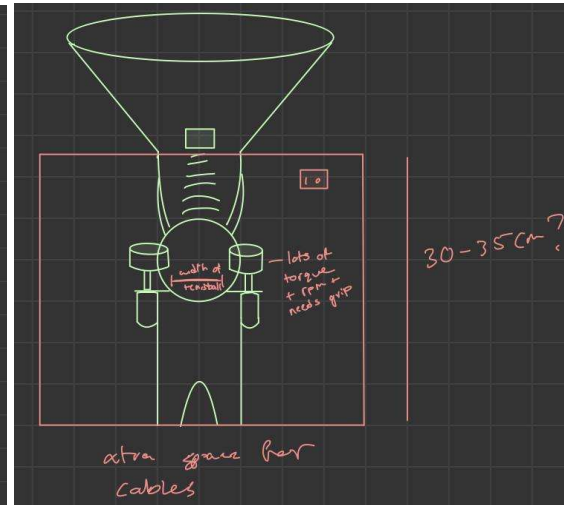
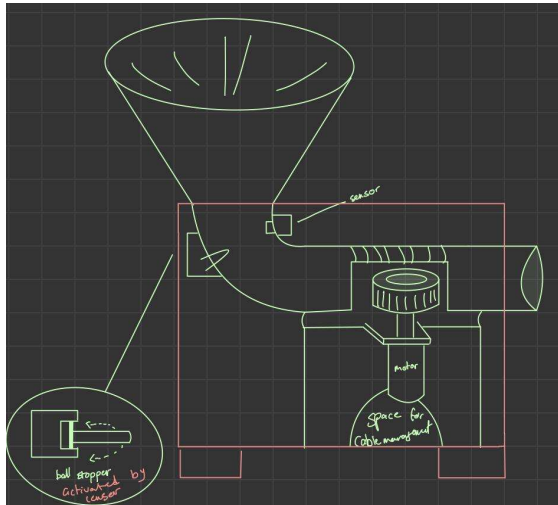
Concept with conveyor belt



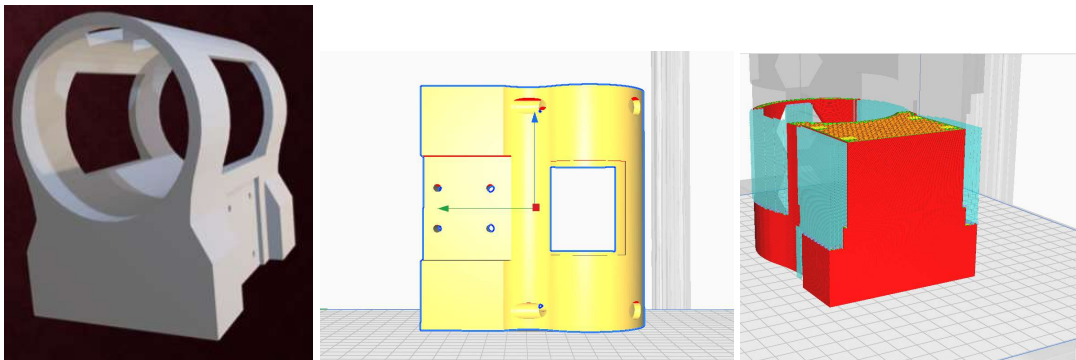
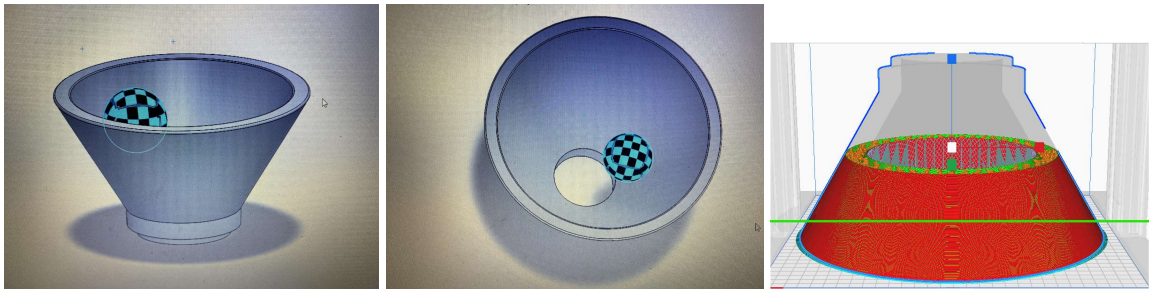
Concept with slope and stopper

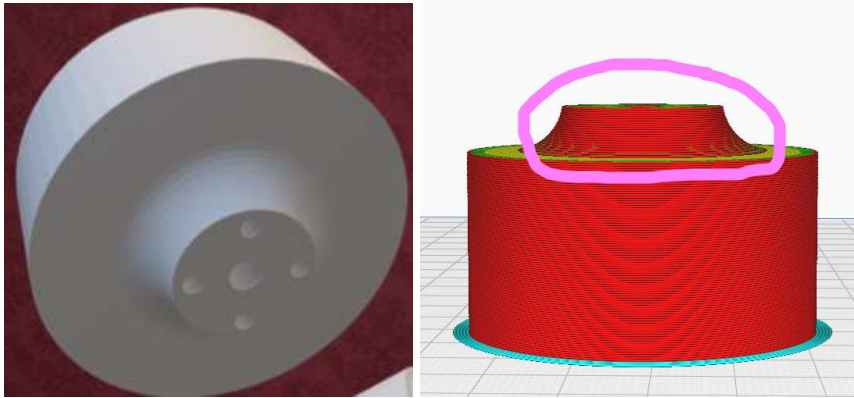


Battery pack schematic

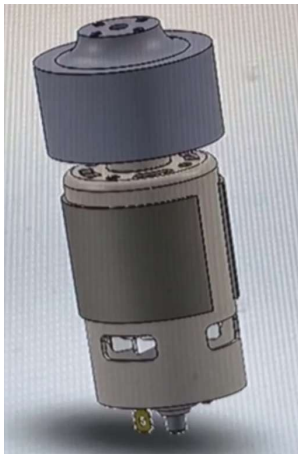


3D models/slicing:





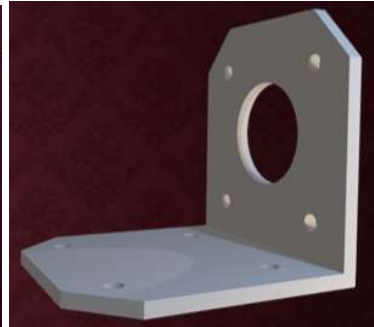
3d model/slicing of the flywheels



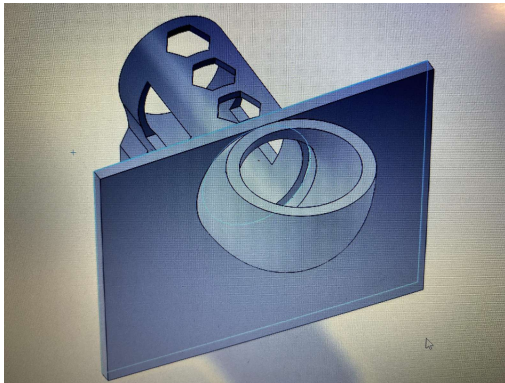
3D model of motor and flywheel



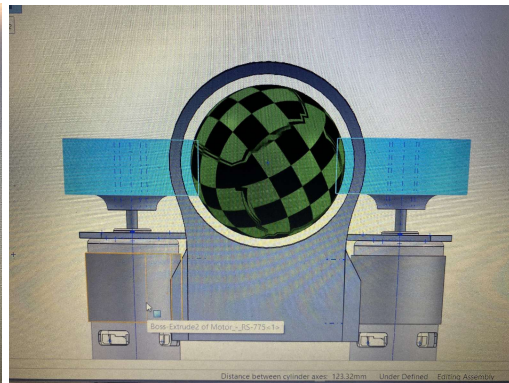
3D model of motor



3D model of the L bracket

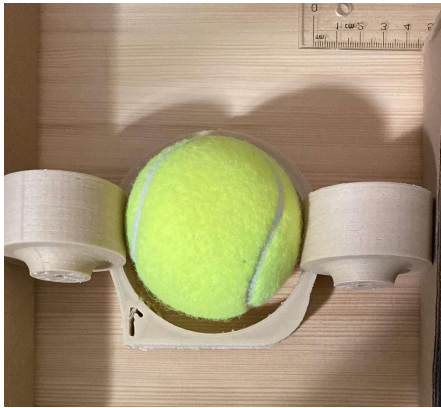


3D model of insert tube

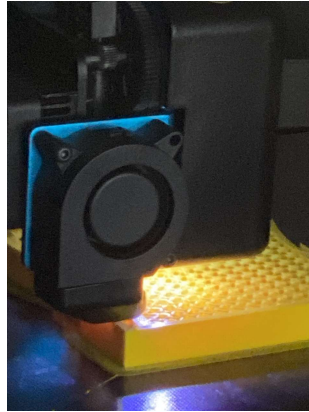


3D model of everything put together

Final product:



Initial fitting test



Printing process



Initial flywheels



Flywheel and coupling



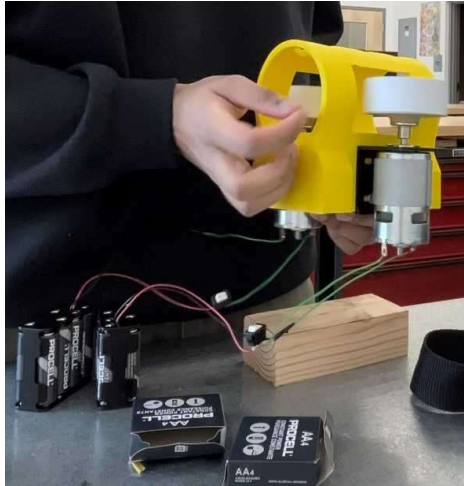
Final flywheels and motors



Partially assembled tube



Fitting test



Fully assembled tube with electrical components



Testing with dogs

Teamworking

As we are near the end of this project we were able to come up with a final version of our project management report:

Get Project Specifications

Before even starting a project having the specs is especially important. From a customer service perspective if you give the customer.

People involved:

Brendan

Generate Concept Solutions/designing

Before building our project we have to generate some concepts. The main idea is to think of problems we could encounter and find solutions to the problems. For example we need to throw a ball well how? During this process we will generate possible solutions to this problem.

People involved

Brendan, Romano, Ahmed, Ryan

Sketch final solution

After generating some concepts we have to narrow it down and find the one solution. This could be a combination of various components from our concepts, or could be something entirely different. The sketches should include drawings and schematics of the final product and every component within, with a listing specifics and notes. The final sketch will start to give us a direction on where to go and lead us to start the manufacturing process.

People involved

Ryan

Research materials

In order to build something you have to have the correct materials. Researching materials is crucial especially if you are on a budget. For example something like carbon fiber can get really expensive because you have to buy things like the fabric, epoxy and other minor things needed. We also have to find out which materials will be the best fit for our final product. This means if a material is cheap but is super fragile to work with it is probably not a good idea to use this material to make our prototype.

People involved

Brendan, Ryan

CAD Drawings

Using school provided software, Solidworks. We will design prototypes using this to have a formal blueprint of how we wish our product will look like.

People involved

Romano, Ahmed

Manufacturing

We need to build our product. This means booking shop times, getting access to special equipment. Getting access to resources. Managing resources and materials under budget. Quality assurance.

People involved.

Brendan, Romano, Ahmed, Ryan

Inspection

This is to make sure that components are up to par and address problems that may cause a disaster and fix them. This also make sure that there were no flaws during the manufacturing process like any cracks in parts

People involved

Ryan, Brendan, Romano

Testing

This is to make sure the product works and meets our standards and to ensure that it functions properly in real life scenarios rather than theoretical scenarios because theoretical results may differ from actual result (ie. tested with real dogs at a dog park)

People involved

Ryan



In summary our roles can be summarized as:

- Brendan Sng: Resources, finance, communications, quality control/manufacturing (3D printing)

- Ryan Thomas: Calculations, electrical, assembly, parts, testing
- Romano Almazan: Solidworks/modeling, schematics, parts
- Ahmed Badawy: Solidworks/modeling, schematics
- We believe that our team was able to fairly distribute the work

We believe that as a team we were able to distribute all the work necessary for this project fairly while also making use of everyone's strengths and circumventing weaknesses. For example, Romano and Ahmed had already completed MSE 100, they were equipped with the required solidworks skills needed to come up with 3D models that we could use to print and were therefore in charge of the main modeling work. Brendan owns and manages his own band, as he has experience in managing groups, we elected him as our product manager. Ryan has plenty of experience in mechanics and electronics as a result of building 4 go karts in high school, as a result he took the main lead in the assembly and testing of the product.

One challenge at the start for us was communication and meeting up. Due to the stress and workload of other courses we were all taking and especially as midterms rolled around we found that when we did do group calls on discord not everyone was able to make it. We solved this problem by assigning specific days and times to meet at the school computer lab and work together there. This is how we found our stride and we noticed that we were way more efficient and productive while all working in person, if one of us had a problem or needed help all of us were there to support. Working in person also made it easier to share ideas, opinions, and to deduce problems to lead to a solution.

Discussion

What were the negative consequences of some of your decisions?